

State of São Paulo ,Brazil population Descriptive_analysis

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R Markdown Notes

1. This is a .rmd document.
2. This is the analysis of 2 datasets.
 - a. **** File**** - 'base_consolidada_sp_2014_2024.parquet'

#	Variable	Type	Description
4	pop_total	integer	Total estimated population
5	pop_masculino	integer	Male population
6	pop_feminino	integer	Female population
7	pop_0_14	integer	Population aged 0–14
8	pop_15_64	integer	Population aged 15–64
9	pop_65plus	integer	Population aged 65 and older

Source: resultados/contextual/populacao_sp_municipio_ano.parquet

- b. populacao_sp_municipio_ano.parquet

Variable	Type	Description
cod_ibge	string	6-digit IBGE municipality code
ano	integer	Reference year
sexo	string	"Masculino" or "Feminino"
idade	integer	Single year of age
populacao	integer	Estimated population count

Source: IBGE intercensal population estimates via DATASUS

3. The Descriptive analysis has the following:
 - Creating quartiles for the population variables and calculating mean, median, range and count.
 - The children's age is divided in 0-3, 4-7,8-10 and 10-14. -Then quartiles were created for each age group grouped by year and calculated mean, median, range and count.
 - Calculating mean, median and range for for population age group 0-14 by sex
 - Proportion of Child Population by Age Group Over Time
 - Percent change in proportion -Plots for propotion variation year wise and sex wise.

Load data

1. Age descriptive analysis

```

pop_data <- pop_data %>%
mutate(
  pct_0_14 = if_else(
    pop_total > 0,
    pop_0_14 / pop_total * 100,
    NA_real_
  ),
  pct_15_64 = if_else(
    pop_total > 0,
    pop_15_64 / pop_total * 100,
    NA_real_
  ),
  pct_65_plus = if_else(
    pop_total > 0,
    pop_65plus / pop_total * 100,
    NA_real_
  )
)
head(pop_data)

```

```

## # A tibble: 6 x 12
##   cod_ibge nome_municipio   ano pop_total pop_masculino pop_feminino pop_0_14
##   <chr>    <chr>          <int>    <int>         <int>         <int>    <int>
## 1 350010 Adamantina (SP)  2014     35150         17016         18134     5992
## 2 350010 Adamantina (SP)  2015     35249         17065         18184     5931
## 3 350010 Adamantina (SP)  2016     35338         17118         18220     5881
## 4 350010 Adamantina (SP)  2017     35415         17154         18261     5843
## 5 350010 Adamantina (SP)  2018     35507         17204         18303     5840
## 6 350010 Adamantina (SP)  2019     35611         17257         18354     5854
## # i 5 more variables: pop_15_64 <int>, pop_65plus <int>, pct_0_14 <dbl>,
## #   pct_15_64 <dbl>, pct_65_plus <dbl>

```

2. Descriptive Statistics of Raw Counts - Quartiles grouped by year

This section summarizes the raw population counts using standard descriptive statistics. These include measures such as:

Quartiles
 Mean population count
 Median population count
 Minimum and maximum values
 Range

```

pop_vars <- c(
  "pop_total",
  "pop_masculino",
  "pop_feminino",
  "pop_0_14",
  "pop_15_64",
  "pop_65plus" )

quartile_summary_by_year <- pop_data %>%
  pivot_longer(
    cols = all_of(pop_vars),
    names_to = "variable",

```

```

    values_to = "value"
  ) %>%
  filter(!is.na(value)) %>%
  group_by(ano, variable) %>%
  mutate(
    Quartile = paste0("Q", ntile(value, 4))
  ) %>%
  group_by(ano, variable, Quartile) %>%
  summarise(
    n = n(),
    Mean = mean(value, na.rm = TRUE),
    Median = median(value, na.rm = TRUE),
    Min = min(value, na.rm = TRUE),
    Max = max(value, na.rm = TRUE),
    .groups = "drop_last"
  ) %>%
  mutate(
    `~` = round(100 * n / sum(n), 2),
    Range = paste0(Min, " - ", Max)
  ) %>%
  ungroup() %>%
  select(
    ano,
    variable,
    Quartile,
    n,
    `~`,
    Mean,
    Median,
    Range
  ) %>%
  arrange(ano, variable, Quartile)

head(quartile_summary_by_year)

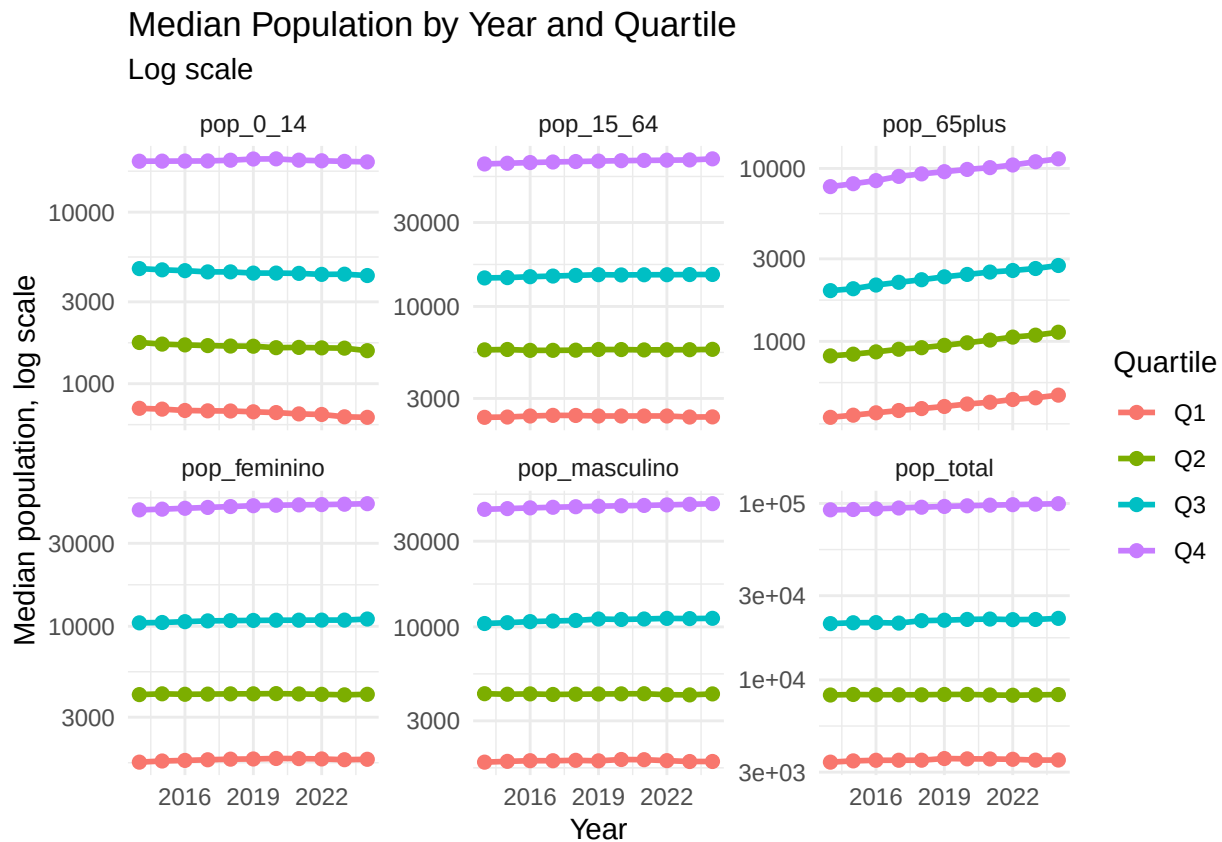
## # A tibble: 6 x 8
##   ano variable  Quartile      n  `~`   Mean Median Range
##   <int> <chr>    <chr>    <int> <dbl> <dbl> <dbl> <chr>
## 1  2014 pop_0_14  Q1      162  25.1   702.   719 164 - 1117
## 2  2014 pop_0_14  Q2      161  25.0  1792.  1739 1139 - 2745
## 3  2014 pop_0_14  Q3      161  25.0  4973.  4688 2753 - 8260
## 4  2014 pop_0_14  Q4      161  25.0 50304. 19805 8291 - 2442916
## 5  2014 pop_15_64 Q1      162  25.1  2329.  2343 583 - 3700
## 6  2014 pop_15_64 Q2      161  25.0  5891.  5667 3707 - 8997

quartile_summary_by_year <- quartile_summary_by_year %>%
  mutate(
    Quartile = factor(Quartile, levels = c("Q1", "Q2", "Q3", "Q4")),
    ano = as.numeric(ano)
  )

#plot
ggplot(quartile_summary_by_year,
  aes(x = ano, y = Median, color = Quartile, group = Quartile)) +

```

```
geom_line(linewidth = 1) +
geom_point(size = 2) +
facet_wrap(~ variable, scales = "free_y") +
scale_y_log10() +
theme_minimal() +
labs(
  title = "Median Population by Year and Quartile",
  subtitle = "Log scale",
  x = "Year",
  y = "Median population, log scale",
  color = "Quartile"
)
```



3. Quartile by splitting the variable pop_0-14

This section summarizes the raw population counts for children of age 0-14 using standard descriptive statistics. These include measures such as:

Quartiles
Mean population count
Median population count
Minimum and maximum values
Range

```
child_data <- read_parquet(
  "/Users/abiramisomasundaram/Desktop/Brazil pesticides/Data/Dados Version 2/Dados/resultados/contextua
")
```

```

child_age_quartiles <- child_data %>%
  filter(idade >= 0, idade <= 14) %>%
  mutate(
    age_group = case_when(
      idade >= 0 & idade <= 3 ~ "0-3",
      idade >= 4 & idade <= 7 ~ "4-7",
      idade >= 8 & idade <= 10 ~ "8-10",
      idade >= 11 & idade <= 14 ~ "11-14",
      TRUE ~ NA_character_
    )
  ) %>%
  filter(!is.na(age_group)) %>%
  group_by(ano, cod_ibge, age_group) %>%
  summarise(
    population = sum(populacao, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  group_by(ano, age_group) %>%
  mutate(
    Quartile = paste0("Q", ntile(population, 4))
  ) %>%
  ungroup()

head(child_age_quartiles)

```

```

## # A tibble: 6 x 5
##   ano  cod_ibge age_group population Quartile
##   <chr> <chr>   <chr>         <int> <chr>
## 1 2014  350010  0-3             1493 Q3
## 2 2014  350010  11-14           1759 Q3
## 3 2014  350010  4-7             1521 Q3
## 4 2014  350010  8-10            1219 Q3
## 5 2014  350020  0-3              189 Q1
## 6 2014  350020  11-14            219 Q1

```

4. Calculating mean, median and range for for population age group 0-14 by year

```

child_quartile_summary <- child_age_quartiles %>%
  group_by(ano, age_group, Quartile) %>%
  summarise(
    n = n(),
    Mean = mean(population, na.rm = TRUE),
    Median = median(population, na.rm = TRUE),
    Min = min(population, na.rm = TRUE),
    Max = max(population, na.rm = TRUE),
    .groups = "drop_last"
  ) %>%
  mutate(
    percent = round(100 * n / sum(n), 2),
    Range = paste0(Min, " - ", Max)
  ) %>%

```

```

ungroup() %>%
select(
  ano,
  age_group,
  Quartile,
  n,
  percent,
  Mean,
  Median,
  Range
) %>%
arrange(ano, age_group, Quartile)

head(child_quartile_summary)

## # A tibble: 6 x 8
##   ano   age_group Quartile      n percent    Mean Median Range
##   <chr> <chr>      <chr>    <int>   <dbl>   <dbl>  <dbl> <chr>
## 1 2014   0-3        Q1      162    25.1    179.   180   45 - 293
## 2 2014   0-3        Q2      161    25.0    462.   442  294 - 709
## 3 2014   0-3        Q3      161    25.0   1285.  1219  715 - 2145
## 4 2014   0-3        Q4      161    25.0  13211.  5256 2155 - 638116
## 5 2014  11-14       Q1      162    25.1    204.   210.  45 - 320
## 6 2014  11-14       Q2      161    25.0    515.   494  324 - 789

```

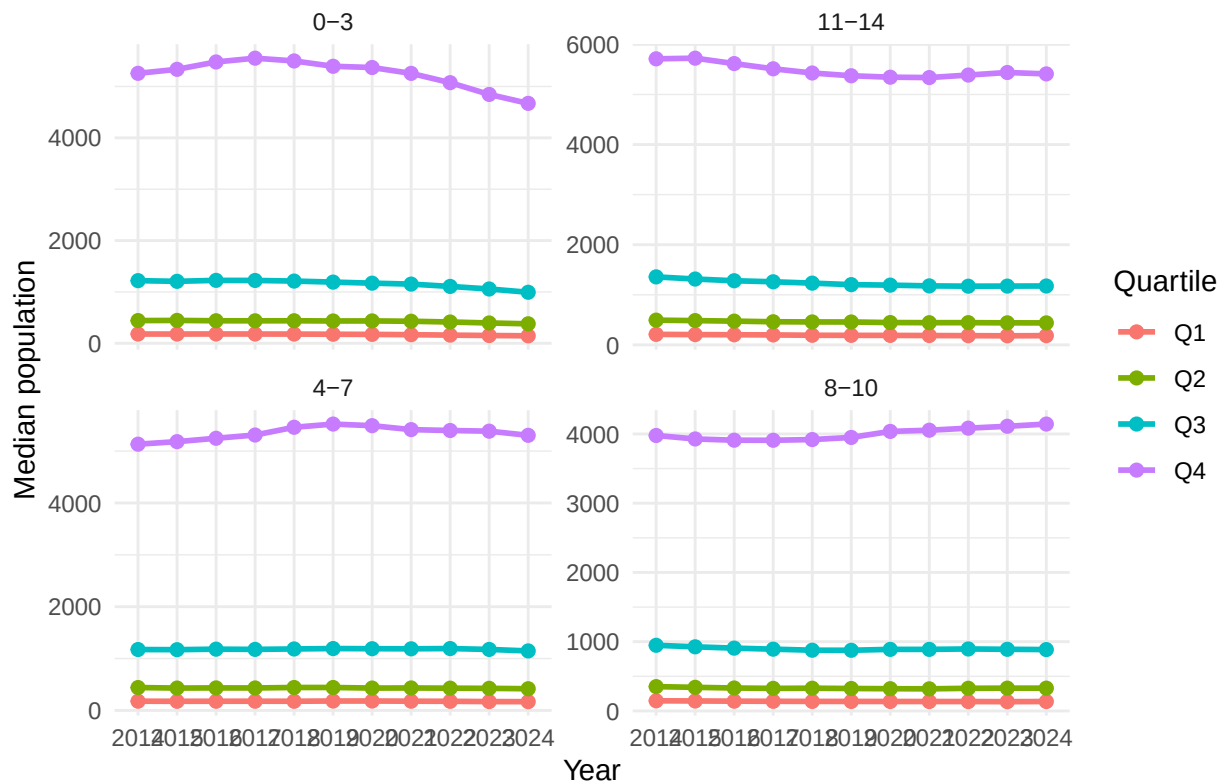
Plot : Median trend over time

```

ggplot(child_quartile_summary,
  aes(x = ano, y = Median, color = Quartile, group = Quartile)) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  facet_wrap(~ age_group, scales = "free_y") +
  theme_minimal() +
  labs(
    title = "Median Child Population by Year and Quartile",
    x = "Year",
    y = "Median population",
    color = "Quartile"
  )

```

Median Child Population by Year and Quartile



5. Calculating mean, median and range for for population age group 0-14 by sex

```
child_age_quartiles_sex <- child_data %>%
  filter(idade >= 0, idade <= 14) %>%
  mutate(
    age_group = case_when(
      idade >= 0 & idade <= 3 ~ "0-3",
      idade >= 4 & idade <= 7 ~ "4-7",
      idade >= 8 & idade <= 10 ~ "8-10",
      idade >= 11 & idade <= 14 ~ "11-14",
      TRUE ~ NA_character_
    )
  ) %>%
  filter(!is.na(age_group)) %>%
  group_by(ano, cod_ibge, sexo, age_group) %>%
  summarise(
    population = sum(populacao, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  group_by(ano, sexo, age_group) %>%
  mutate(
    Quartile = paste0("Q", ntile(population, 4))
  ) %>%
  ungroup()
```

```

child_quartile_summary_sex <- child_age_quartiles_sex %>%
  group_by(ano, sexo, age_group, Quartile) %>%
  summarise(
    n = n(),
    Mean = mean(population, na.rm = TRUE),
    Median = median(population, na.rm = TRUE),
    Min = min(population, na.rm = TRUE),
    Max = max(population, na.rm = TRUE),
    .groups = "drop_last"
  ) %>%
  mutate(
    percent = round(100 * n / sum(n), 2),
    Range = paste0(Min, " - ", Max)
  ) %>%
  ungroup() %>%
  arrange(ano, sexo, age_group, Quartile)

head(child_quartile_summary_sex)

```

```

## # A tibble: 6 x 11
##   ano  sexo  age_group Quartile      n  Mean Median   Min    Max percent Range
##   <chr> <chr>   <chr>    <chr>   <int> <dbl> <dbl> <int> <int>   <dbl> <chr>
## 1 2014  Femin~  0-3      Q1      162  87.4    86    21    144    25.1 21  ~
## 2 2014  Femin~  0-3      Q2      161  226.   218   144    350    25.0 144 ~
## 3 2014  Femin~  0-3      Q3      161  627.   600   350   1047    25.0 350 ~
## 4 2014  Femin~  0-3      Q4      161 6448.  2555  1062  311551  25.0 1062~
## 5 2014  Femin~ 11-14     Q1      162  98.1   102    21    155    25.1 21  ~
## 6 2014  Femin~ 11-14     Q2      161  252.   239   155    381    25.0 155 ~

```

#6. Create age groups and calculation of yearly proportions(%)

```

child_age_yearly <- child_data %>%
  filter(idade >= 0, idade <= 14) %>%
  mutate(
    age_group = case_when(
      idade >= 0 & idade <= 3 ~ "0-3",
      idade >= 4 & idade <= 7 ~ "4-7",
      idade >= 8 & idade <= 10 ~ "8-10",
      idade >= 11 & idade <= 14 ~ "11-14",
      TRUE ~ NA_character_
    )
  ) %>%
  filter(!is.na(age_group)) %>%
  group_by(ano, age_group) %>%
  summarise(
    total_population = sum(populacao, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  group_by(ano) %>%
  mutate(
    total_child_population = sum(total_population, na.rm = TRUE),
    proportion = total_population / total_child_population,
    proportion_percent = 100 * proportion
  ) %>%

```

```

ungroup() %>%
  arrange(age_group, ano)

head(child_age_yearly)

## # A tibble: 6 x 6
##   ano   age_group total_population total_child_population proportion
##   <chr> <chr>          <int>          <int>          <dbl>
## 1 2014   0-3            2437108            9301971        0.262
## 2 2015   0-3            2459419            9253638        0.266
## 3 2016   0-3            2460587            9204409        0.267
## 4 2017   0-3            2449433            9153990        0.268
## 5 2018   0-3            2435760            9112558        0.267
## 6 2019   0-3            2399633            9065871        0.265
## # i 1 more variable: proportion_percent <dbl>

```

7. Percent change in proportion

```

child_age_variation <- child_age_yearly %>%
  group_by(age_group) %>%
  arrange(ano, .by_group = TRUE) %>%
  mutate(
    previous_proportion_percent = lag(proportion_percent),

    percentage_point_change = proportion_percent - previous_proportion_percent,

    percent_change_in_proportion = 100 *
      (proportion_percent - previous_proportion_percent) / previous_proportion_percent
  ) %>%
  ungroup()

head(child_age_variation)

## # A tibble: 6 x 9
##   ano   age_group total_population total_child_population proportion
##   <chr> <chr>          <int>          <int>          <dbl>
## 1 2014   0-3            2437108            9301971        0.262
## 2 2015   0-3            2459419            9253638        0.266
## 3 2016   0-3            2460587            9204409        0.267
## 4 2017   0-3            2449433            9153990        0.268
## 5 2018   0-3            2435760            9112558        0.267
## 6 2019   0-3            2399633            9065871        0.265
## # i 4 more variables: proportion_percent <dbl>,
## #   previous_proportion_percent <dbl>, percentage_point_change <dbl>,
## #   percent_change_in_proportion <dbl>

```

8. Proportion of Child Population by Age Group Over Time

This shows the percentage share of each age group each year.

```

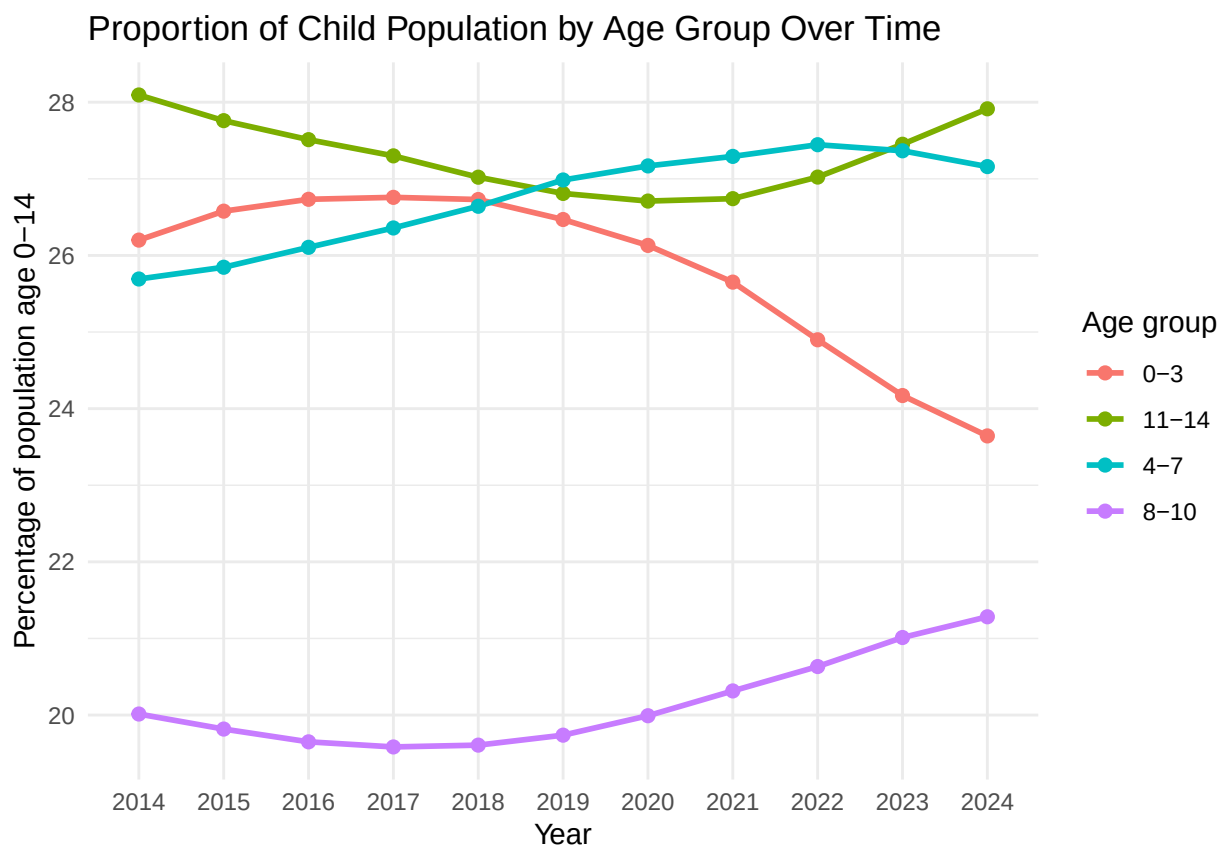
ggplot(
  child_age_variation,
  aes(

```

```

    x = ano,
    y = proportion_percent,
    color = age_group,
    group = age_group
  )
) +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  labs(
    title = "Proportion of Child Population by Age Group Over Time",
    x = "Year",
    y = "Percentage of population age 0-14",
    color = "Age group"
  ) +
  theme_minimal()

```



#8. Proportional variation year-wise

```

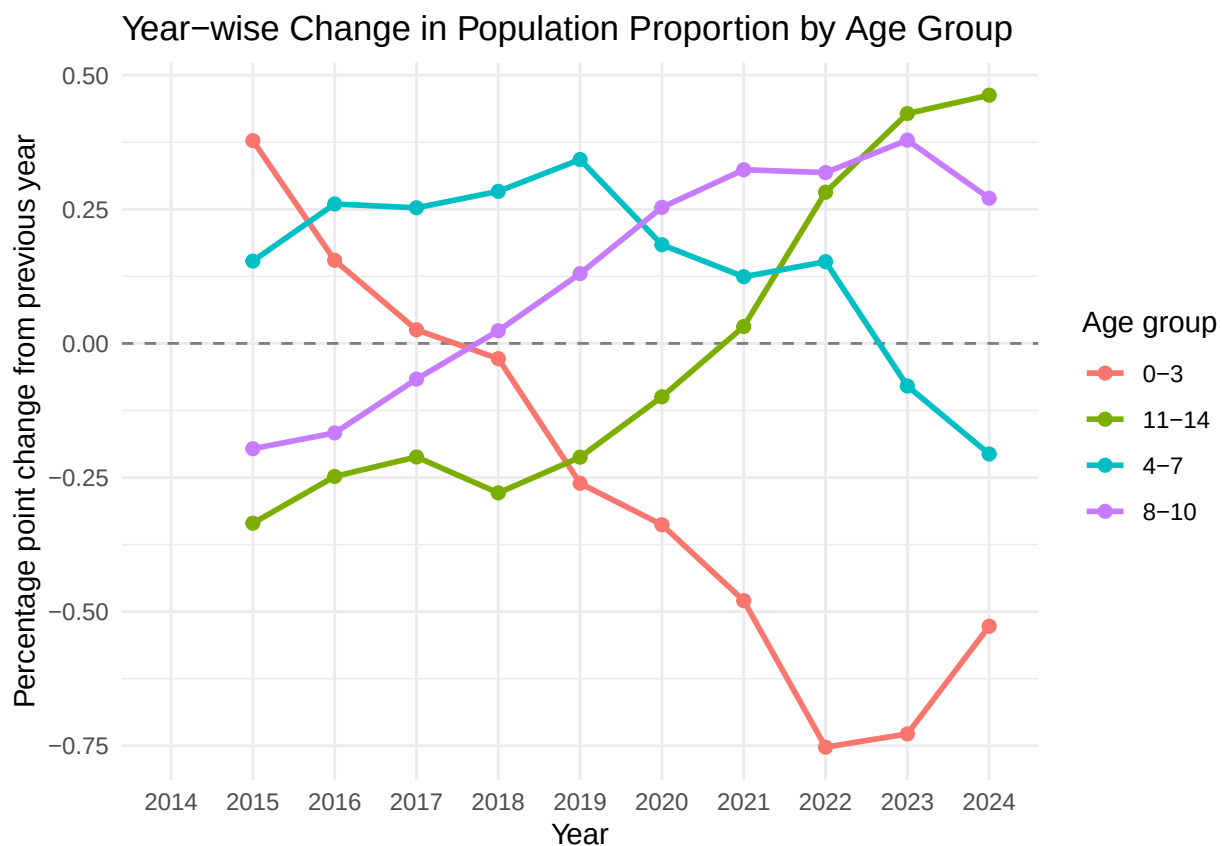
ggplot(
  child_age_variation,
  aes(
    x = ano,
    y = percentage_point_change,
    color = age_group,
    group = age_group
  )
) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +

```

```
geom_line(linewidth = 1) +
geom_point(size = 2) +
labs(
  title = "Year-wise Change in Population Proportion by Age Group",
  x = "Year",
  y = "Percentage point change from previous year",
  color = "Age group"
) +
theme_minimal()
```

```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## (`geom_line()`).
```

```
## Warning: Removed 4 rows containing missing values or values outside the scale range
## (`geom_point()`).
```



```
child_age_yearly_sex <- child_data %>%
  filter(idade >= 0, idade <= 14) %>%
  mutate(
    age_group = case_when(
      idade >= 0 & idade <= 3 ~ "0-3",
      idade >= 4 & idade <= 7 ~ "4-7",
      idade >= 8 & idade <= 10 ~ "8-10",
      idade >= 11 & idade <= 14 ~ "11-14",
      TRUE ~ NA_character_
    )
  ) %>%
  filter(!is.na(age_group)) %>%
```

```

group_by(ano, sexo, age_group) %>%
  summarise(
    total_population = sum(populacao, na.rm = TRUE),
    .groups = "drop"
  ) %>%
  group_by(ano, sexo) %>%
  mutate(
    total_child_population = sum(total_population, na.rm = TRUE),
    proportion_percent = 100 * total_population / total_child_population
  ) %>%
  ungroup() %>%
  arrange(sexo, age_group, ano)

child_age_variation_sex <- child_age_yearly_sex %>%
  group_by(sexo, age_group) %>%
  arrange(ano, .by_group = TRUE) %>%
  mutate(
    previous_proportion_percent = lag(proportion_percent),
    percentage_point_change = proportion_percent - previous_proportion_percent,
    percent_change_in_proportion = 100 *
      (proportion_percent - previous_proportion_percent) / previous_proportion_percent
  ) %>%
  ungroup()

#Plot
ggplot(
  child_age_variation_sex,
  aes(
    x = ano,
    y = percentage_point_change,
    color = age_group,
    group = age_group
  )
) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  geom_line(linewidth = 1) +
  geom_point(size = 2) +
  facet_wrap(~ sexo) +
  labs(
    title = "Year-wise Change in Child Population Proportion by Age Group and Sex",
    x = "Year",
    y = "Percentage point change from previous year",
    color = "Age group"
  ) +
  theme_minimal()

```

```

## Warning: Removed 8 rows containing missing values or values outside the scale range
## (`geom_line()`).

```

```

## Warning: Removed 8 rows containing missing values or values outside the scale range
## (`geom_point()`).

```

Year-wise Change in Child Population Proportion by Age Group and Sex

